Trainings

Select Service Tools

Recommended Cummins® Service Tools

- Oil carryover test tool, Part Number 4918883 (no longer available)
- Cummins® electronic service tool, or equivalent

Additional Service Items

- High-temperature flexible hose.

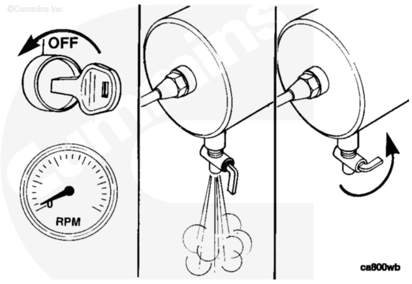
Inspect for Reuse

Perform this test in conjunction with troubleshooting the air compressor pumping oil into the air system.

Note : Small oil deposits at the air dryer purge valve are normal. The air compressor is lubricated with engine oil, and small amounts of carryover are to be expected. Oil carryover is more common on naturally aspirated air compressors. Oil carryover is usually caused by high inlet restriction to the air compressor. If oil carryover is suspected or found, check the engine air filter and air supply plumbing. See equipment manufacturer service information.

⚠ WARNING ⚠

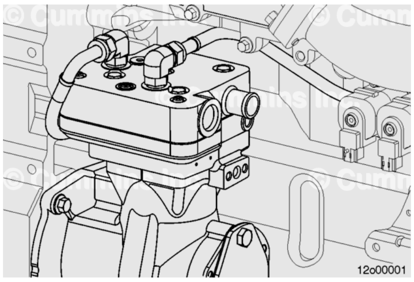
The air discharge line and other equipment will become hot during the course of the test. To prevent burns, use protective gloves when touching heated surfaces.



Start the engine and run until the coolant temperature reaches normal operating temperature. Once the coolant has reached operating temperature, shut the engine down and completely drain the vehicle air system.

⚠ WARNING ⚠

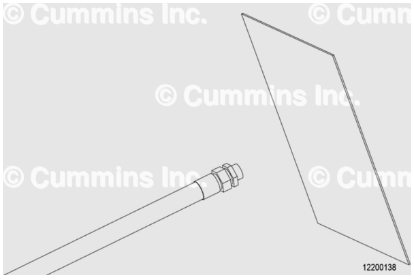
The discharge from the air compressor delivery will be hot, possibly contain oil vapors, and will be noisy. Be sure that there is adequate ventilation and that hearing protection is used, particularly if the type of equipment requires that the test be conducted in an enclosed environment.



Disconnect the discharge pipe from the air compressor cylinder head.

Disconnect the air inlet plumbing at the air compressor cylinder head.

Connect the oil carryover test tool, Part Number 4918883, to the air compressor cylinder head discharge port. This tool has been discontinued and is no longer available for purchase. If you do **not** have the oil carryover test tool, Part Number 4918883, remove the air compressor discharge line from the air dryer, and position the test paper at a distance of 75 mm [3.0 in], **not** to exceed 100 mm [3.9 in], from the end of the Original Equipment Manufacturer (OEM) compressor discharge line or flexible hose if fitted.



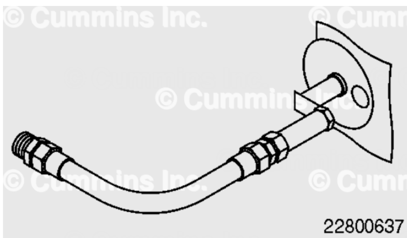
Note : If the application presents restricted access to the air compressor, a high-temperature flexible hose, minimum 15 mm [0.6 in] inside diameter, may be connected directly to the discharge port of the compressor to carry out the test outside the engine compartment.

Start the engine and operate at high idle for 5 minutes to stabilize the temperature.

Once the engine operating temperature is reached, shut the engine down.

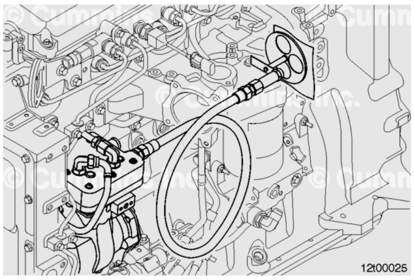
Note : It may be necessary to use the recommended Cummins® electronic service tool, or equivalent, to increase the maximum engine speed, without Vehicle Speed Sensor (VSS), to rated speed.

Install the test paper into the oil carryover test tool, Part Number 4918883, if available. Make sure it is held at a right angle to and in line with the flow of compressed air, at a distance **not** to exceed 100 mm [3.9 in] from the end of the compressor discharge pipe, or flexible hose, if fitted.



The test paper can be standard clean copier paper, typically 80 grams/sq meter. It **must** be mounted in the air compressor discharge line tool, leaving a 70 mm [2.76 in] diameter circle of the paper exposed. The outline of the circle in the mask **must** be drawn onto the test paper to later aid in comparison.

To perform the test, start the engine and run at high idle. Leave the test paper in the airflow for exactly 5 minutes. Remove the test paper, or rotate the tool mask exposing a new circular test area.

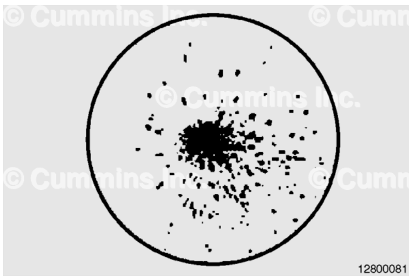


Repeat this test three times, using a new circular test area/paper each time.

Shut the engine off.

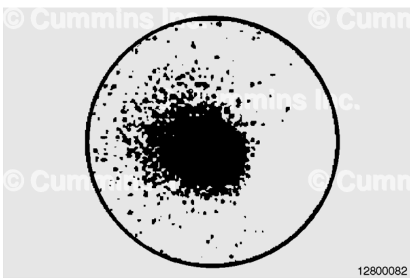
Note : If the maximum engine speed, without VSS, is adjusted in the previous step, change it back to the original value upon completion of the test.

Note : Compare the test results with the reference results shown below.



Time Air Compressor Has Been In Service (Months/Years):

- Less than 6 months, 62,500 miles, 100,584 kilometers, or 1563 hours
  - Compressor is in good condition.
- 6 months to 2 years, 62,500 to 250,000 miles, 100,584 to 402,336 kilometers, or 1563 to 6250 hours
  - Compressor is in good condition.
- Greater than 2 years, 250,000 miles, 402,336 kilometers, or 6250 hours
  - Compressor is in good condition.

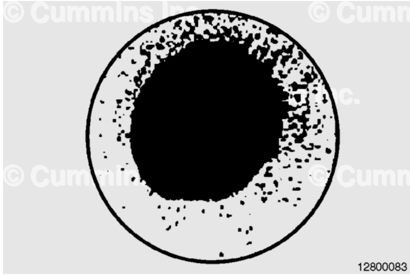


Time Air Compressor Has Been In Service (Months/Years):

- Less than 6 months, 62,500 miles, 100,584 kilometers, or 1563 hours
  - Compressor needs to be replaced.
- 6 months to 2 years, 62,500 to 250,000 miles, 100,584 to 402,336 kilometers, or 1563 to 6250 hours
  - Compressor is in acceptable condition.
- Greater than 2 years, 250,000 miles, 402,336 kilometers, or 6250 hours
  - Compressor is in good condition.

Time Air Compressor Has Been In Service (Months/Years):

- Less than 6 months, 62,500 miles, 100,584 kilometers, or 1563 hours
  - Compressor needs to be replaced.
- 6 months to 2 years, 62,500 to 250,000 miles, 100,584 to 402,336 kilometers, or 1563 to 6250 hours
  - Compressor needs to be replaced.
- Greater than 2 years, 250,000 miles, 402,336 kilometers, or 6250 hours
  - Compressor needs to be replaced.



Note : This step must be completed if the compressor assembly is replaced as a result of this test and is requested to be returned via the CORE or enhanced parts return process.

If the compressor is determined to be malfunctioning and needs to be replaced, place the test paper with the results of the Oil Carryover Test in a sealable plastic bag, seal to prevent oil contamination, and include with the malfunctioning compressor.

Last Modified: 17-Nov-2022